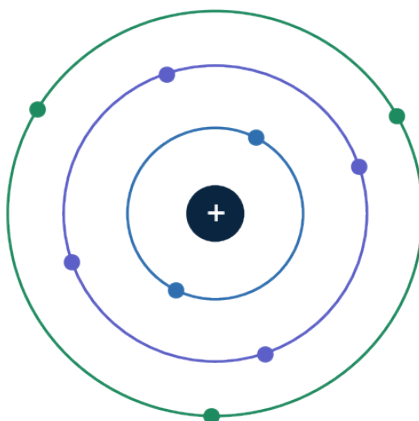


Atomic Structure

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Bohr's model: electrons revolve in fixed circular orbits (shells) without radiating energy; energy is absorbed/emitted only when an electron jumps between orbits.
- Energy of an electron in the n th orbit of a hydrogen-like atom: $E_n = -13.6 Z^2 / n^2 \text{ eV}$.
- Quantum numbers describe an electron's state: principal (n , shell/energy), azimuthal (l , subshell/shape, 0 to $n-1$), magnetic (m_l , orientation), spin (m_s , $+1/2$ or $-1/2$).
- Aufbau principle: electrons fill orbitals in order of increasing energy (1s, 2s, 2p, 3s, 3p, 4s, 3d, ...).
- Pauli's exclusion principle: no two electrons in an atom can have the same set of all four quantum numbers.
- Hund's rule: electrons fill degenerate orbitals singly (with parallel spin) before pairing up.
- de Broglie wavelength: $\lambda = h / (mv)$, showing the wave nature of matter. Heisenberg's uncertainty principle: $\Delta x \cdot \Delta p \geq h/(4\pi)$.



Bohr model: electrons occupy discrete shells around a central nucleus.

2. Formula Sheet

Bohr orbit energy (H-like atom)	$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$
Bohr orbit radius	$r_n = 0.529 \frac{n^2}{Z} \text{ Angstrom}$
Rydberg formula	$\frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$
de Broglie wavelength	$\lambda = h / (m v)$
Heisenberg uncertainty principle	$\Delta x \cdot \Delta p \geq h / (4 \pi)$
Max electrons in a shell	$2 n^2$
Max electrons in a subshell	$2(2l + 1)$
Planck's relation	$E = h \nu = hc / \lambda$

3. Practice Problems (22 questions)

- Find the energy of an electron in the 2nd orbit of a hydrogen atom ($Z=1$).
- Find the radius of the 1st Bohr orbit of hydrogen.
- Find the maximum number of electrons that can be accommodated in the $n=3$ shell.
- Write the electronic configuration of chlorine ($Z=17$).
- State the four quantum numbers and what each one describes.
- Find the de Broglie wavelength of an electron (mass $9.1 \times 10^{-31} \text{ kg}$) moving at 10^6 m/s ($h = 6.63 \times 10^{-34} \text{ J s}$).
- State Hund's rule and use it to write the electron arrangement in the 2p subshell for nitrogen ($Z=7$).
- Find the number of subshells possible in the $n=4$ shell.
- State the Pauli exclusion principle and explain why it limits each orbital to 2 electrons.
- Find the wavelength of light emitted when an electron in a hydrogen atom falls from $n=3$ to $n=2$ ($R = 1.097 \times 10^7 / \text{m}$).
- Explain the difference between an orbit (Bohr model) and an orbital (quantum mechanical model).
- Find the energy required to ionize a hydrogen atom from its ground state ($n=1$).
- Write the electronic configuration of Ca^{2+} (Ca , $Z=20$).
- State the Aufbau principle and explain one exception to the simple filling order (e.g., Cr or Cu).
- Find the number of radial nodes in the 3s orbital.
- Explain why the energy of electrons in a multi-electron atom depends on both n and l , not just n .
- Find the frequency of a photon with energy $3.03 \times 10^{-19} \text{ J}$ ($h = 6.63 \times 10^{-34} \text{ J s}$).
- State Heisenberg's uncertainty principle and explain its significance for the concept of a fixed electron orbit.
- Find the number of unpaired electrons in the ground state electronic configuration of iron (Fe , $Z=26$).

20. Explain why the Bohr model successfully explains the hydrogen spectrum but fails for multi-electron atoms.
21. Find the maximum number of electrons in an orbital with $l=2$ (d subshell).
22. State the significance of the azimuthal quantum number (l) and list the subshell notations for $l=0,1,2,3$.
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Answer Key

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|---|--|
| 1. $E_2 = -3.4$ eV | 2. $r_1 = 0.529$ Angstrom |
| 3. 18 electrons | 4. $1s^2 2s^2 2p^6 3s^2 3p^5$ |
| 5. Conceptual - n (shell/size), l (subshell/shape), m_l (orientation), m_s (spin) | 6. $\lambda \approx 7.27 \times 10^{-10}$ m |
| 7. Conceptual - 3 unpaired electrons in 2p, each in a separate orbital with parallel spin (Hund's rule) | 8. 4 subshells (4s, 4p, 4d, 4f) |
| 9. Conceptual - each electron has a unique quantum state due to opposite spins in the same orbital | 10. $\lambda \approx 6.56 \times 10^{-7}$ m (656 nm) |
| 11. Conceptual - orbit is a fixed classical path; orbital is a probability region (3D electron cloud) with no definite path | 12. 13.6 eV |
| 13. $1s^2 2s^2 2p^6 3s^2 3p^6$ (same as Argon configuration) | 14. Conceptual - electrons fill in order of increasing energy; Cr and Cu show exceptions due to extra stability of half-filled/filled d-subshells |
| 15. 2 radial nodes ($n-l-1 = 3-0-1 = 2$) | 16. Conceptual - in multi-electron atoms, shielding and penetration effects make energy depend on both n and l |
| 17. $\nu \approx 4.57 \times 10^{14}$ Hz | 18. Conceptual - position and momentum cannot both be precisely known simultaneously, so a fixed circular orbit (as in Bohr model) is not physically valid |
| 19. 4 unpaired electrons | 20. Conceptual - Bohr model works for one-electron systems (H, He^+) but ignores electron-electron repulsions present in multi-electron atoms |
| 21. 10 electrons ($2(2 \times 2 + 1) = 10$) | 22. Conceptual - $l = 0(s), 1(p), 2(d), 3(f)$; determines subshell shape |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.